

Collision Geometry of Joint Spectra

Affine Branch Arrangements, Exact Finite Censuses, and Conditional Spectral Quotients

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Abstract

Problem. A finite joint spectrum can be observed at two resolutions: a resolved set of joint spectral points and a coarser spectrum obtained by a linear functional. We ask how collisions of affine branches organize the resulting quotient, explicitly separating general finite-point theorems from a numerically registered operator realization.

Approach. For a fixed finite arrangement $P = \{(q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2$, we study $L_\alpha(q, h) = \alpha q + (1 - \alpha)h$ and the equivalence relation induced by equal L_α values. We prove the collision formula, finiteness of the critical set, constancy of branch order on its complementary chambers, and the spectral quotient theorem for commuting Hermitian realizations. Rational coordinates give rational finite collision parameters. We then perform a complete exact census for one explicitly declared nine-point rational arrangement.

Results. For the explicit arrangement P_9 , the 36 unordered pairs split into 2 parallel, 10 interior, 15 endpoint, and 9 exterior pairs. On $[0, 1]$ the interior critical parameters are $2/7$, $2/5$, $1/2$, $2/3$, and $4/5$. The layer-count drops are respectively 1, 2, 2, 3, and 1, so $\alpha = 2/3$ is the unique interior parameter of maximal drop. Its nontrivial quotient classes contain S5, S6, and S7, and S8 and S9, respectively.

Boundary. In the declared complex128 Rubik computation, coordinate-matched labels, ranks, and projectors are stable across the tested clustering tolerances, and the maximum raw-to- P_9 table discrepancy is below 10^{-15} . Machine-zero commutator and projector residuals support this registration but do not prove exact commutation or exact rational joint eigenvalues. Conditional on exact registration, the six averaging layers are the $L_{2/3}$ collision quotient of P_9 . No parameter-dependent or maximality conclusion is drawn.

Notation and Claim Layers

Symbol	Meaning
$P = \{p_i\}_{i=1}^N$	fixed finite set of distinct points $p_i = (q_i, h_i)$
$L_\alpha(q, h)$	$\alpha q + (1 - \alpha)h$
$\lambda_i(\alpha)$	affine branch $L_\alpha(p_i)$
$\mathcal{C}(P)$	finite collision set in $\mathbb{R}$
$i \sim_\alpha j$	equality $L_\alpha(p_i) = L_\alpha(p_j)$
$P / \sim_\alpha$	collision quotient at α
$N_P(\alpha)$	number of quotient classes
$\Delta_P(\alpha)$	layer-count drop $ P - N_P(\alpha)$
$m_{\text{pair}}(\alpha)$	number of colliding unordered pairs at α
P_9	explicit weighted nine-point rational arrangement
Q, H	commuting Hermitian operators in the exact general theory
$Q_{\text{num}}, H_{\text{num}}$	numerically constructed Rubik QT/HT averages
A_{18}	canonical 18-generator Rubik average

The paper uses four claim-status levels:

1. **Theorem:** general finite-point and commuting-Hermitian statements, the exact finite census of P_9 , and the conditional Rubik corollary;
 2. **Computational Certificate:** the declared operator-to- P_9 registration and residual audit;
 3. **Computational Observation:** finite transport comparisons that do not supply a promotion theorem;
 4. **Research Program:** statements explicitly identified as open and not used in the proofs.
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1 Introduction

Let a fixed finite joint-spectrum arrangement be given by

$$P = \{(q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2.$$

The one-parameter family of linear observables

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h$$

assigns to each point an affine branch

$$\lambda_i(\alpha) = L_\alpha(q_i, h_i).$$

At a generic parameter the branch values are distinct. At a critical parameter, two or more values coincide and the resolved point set is replaced by a quotient partition. If P is the exact joint spectrum of commuting Hermitian operators Q and H , this quotient is precisely the eigenspace decomposition of $A(\alpha) = \alpha Q + (1 - \alpha)H$.

This observation is elementary, but three logically distinct tasks must not be conflated:

- proving the general finite-point quotient theorem;
- classifying one explicitly declared rational arrangement exactly;
- showing that a numerical operator computation realizes that arrangement.

The present paper separates these tasks. Its contributions are:

1. a self-contained collision-quotient theorem for fixed finite arrangements;
2. a rationality proposition for finite collision parameters;
3. a complete exact phase census on $[0, 1]$ for a weighted nine-point rational arrangement P_9 ;
4. a claim-status-aware numerical registration of the Rubik QT/HT clusters;
5. a conditional corollary identifying the Rubik averaging layers with the exact quotient when the registration is promoted to exact data.

The general theory does not require Rubik data. Conversely, the numerical registration does not become exact merely because the downstream finite-point arithmetic uses rational numbers. This direction of dependence is the central claim discipline of the paper.

The fixed-arrangement output may be applied pointwise once an exact or explicitly registered joint spectrum is available. By contrast, any parameter-dependent application requires a certified normal spectral chart and coherent projector continuation.

2 Finite Collision Arrangements

Definition 2.1 (Affine branch arrangement).

Let $P = \{p_i = (q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2$ be finite with distinct points. For $\alpha \in \mathbb{R}$, define

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h, \quad \lambda_i(\alpha) = L_\alpha(p_i).$$

Thus

$$\lambda_i(\alpha) = h_i + \alpha(q_i - h_i).$$

Definition 2.2 (Collision set and quotient).

The collision set is

$$\mathcal{C}(P) = \{\alpha \in \mathbb{R} : \lambda_i(\alpha) = \lambda_j(\alpha) \text{ for some } i \neq j\}.$$

At fixed α , define

$$i \sim_\alpha j \iff \lambda_i(\alpha) = \lambda_j(\alpha).$$

The partition $P/\sim_\alpha$ is the collision quotient. Write

$$N_P(\alpha) = |P/\sim_\alpha|, \quad \Delta_P(\alpha) = N - N_P(\alpha).$$

Definition 2.3 (Pair multiplicity and quotient drop).

Let $C_1, \dots, C_r$ be the non-singleton classes of $P/\sim_\alpha$. Define

$$m_{\text{pair}}(\alpha) = \sum_{a=1}^r \binom{|C_a|}{2}, \quad \Delta_P(\alpha) = \sum_{a=1}^r (|C_a| - 1).$$

These two quantities are distinct. For example, a triple collision contributes three pair equalities but only two units of layer-count drop.

Definition 2.4 (Branch order).

For $\alpha \notin \mathcal{C}(P)$, the values $\lambda_i(\alpha)$ are distinct. Their descending order defines the branch order at α . A chamber is a connected component of $\mathbb{R} \setminus \mathcal{C}(P)$.

The quotient partition is discrete on every chamber. The branch order is the additional datum that distinguishes neighboring chambers.

3 General Collision–Quotient Theorem

Theorem 3.1 (Finite collision arrangement).

Let $P = \{(q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2$ be a finite set of distinct points.

1. If the branches i and j have different slopes, they collide at the unique parameter

$$\alpha_{ij} = \frac{h_j - h_i}{(q_i - h_i) - (q_j - h_j)}.$$

2. If the slopes agree, the distinct branches are parallel and never collide.
3. The collision set $\mathcal{C}(P)$ is finite.

4. On each chamber of $\mathbb{R} \setminus \mathcal{C}(P)$, the quotient is discrete and the strict branch order is constant.
5. At a critical parameter, the non-singleton collision classes are exactly the connected components of the graph whose edges are the pairs satisfying $\alpha_{ij} = \alpha$.

Proof. Equality of two branches gives

$$h_i + \alpha(q_i - h_i) = h_j + \alpha(q_j - h_j).$$

If the coefficient of α is nonzero, solving for α yields the displayed formula and guarantees uniqueness. If the coefficient is zero, the slopes agree. The distinctness of the points then implies $h_i \neq h_j$, so the parallel affine functions never meet. Because there are only $\binom{N}{2}$ pairs, the collision set is finite.

Outside $\mathcal{C}(P)$, every difference $\lambda_i - \lambda_j$ is nonzero. Its sign is continuous and therefore constant on each connected chamber. Consequently, the strict branch order remains constant within each chamber. At a critical parameter, equality is transitive; thus, the equality classes correspond exactly to the connected components of the equal-label graph.

Theorem 3.2 (Spectral layers as collision classes).

Let Q, H be exactly commuting Hermitian operators on a finite-dimensional complex Hilbert space. Let

$$V = \bigoplus_{i=1}^N E_i, \quad Q|_{E_i} = q_i I, \quad H|_{E_i} = h_i I,$$

where equal joint eigenvalue pairs have already been combined. Then the eigenspaces of

$$A(\alpha) = \alpha Q + (1 - \alpha)H$$

are precisely

$$\bigoplus_{i \in C} E_i, \quad C \in P / \sim_\alpha.$$

Proof. On E_i , the operator $A(\alpha)$ acts by the scalar $L_\alpha(q_i, h_i)$. Two joint eigenspaces belong to the same eigenspace of $A(\alpha)$ exactly when their scalar values agree.

Boundary. This theorem assumes exact commutation and exact joint spectral data. A small numerical commutator residual does not establish its hypotheses.

4 Rational Arrangements

Proposition 4.1 (Rational collision parameters).

If $P \subset \mathbb{Q}^2$ is finite, every finite collision parameter belongs to $\mathbb{Q}$.

Proof. The collision formula uses subtraction and division by a nonzero rational number.

We do not claim a denominator bound that depends solely on a common coordinate denominator. Such a bound would also require a declared input height or coordinate range. For the explicit arrangement studied below, small denominators are established through a direct exact census rather than by invoking a general height theorem. The comparison with Farey-type small-denominator organization is descriptive only [7].

5 Exact Nine-Point Census

Define the following weighted rational arrangement independently of any operator realization:

$$P_9 = \{(S_i, d_i, q_i, h_i) : 1 \leq i \leq 9\}.$$

Label	Weight d_i	q_i	h_i	$q_i - h_i$	$L_{2/3}(q_i, h_i)$
S1	20	1	1	0	1
S2	2	5/6	1	-1/6	8/9
S3	39	5/6	2/3	1/6	7/9
S4	26	1/2	1	-1/2	2/3
S5	1	1/3	1	-2/3	5/9
S6	39	1/2	2/3	-1/6	5/9
S7	66	2/3	1/3	1/3	5/9
S8	8	0	1	-1	1/3
S9	27	1/3	1/3	0	1/3

The weights sum to 228. In this section, they are attached integer weights; their interpretation as operator ranks belongs to the computational registration and conditional corollary below.

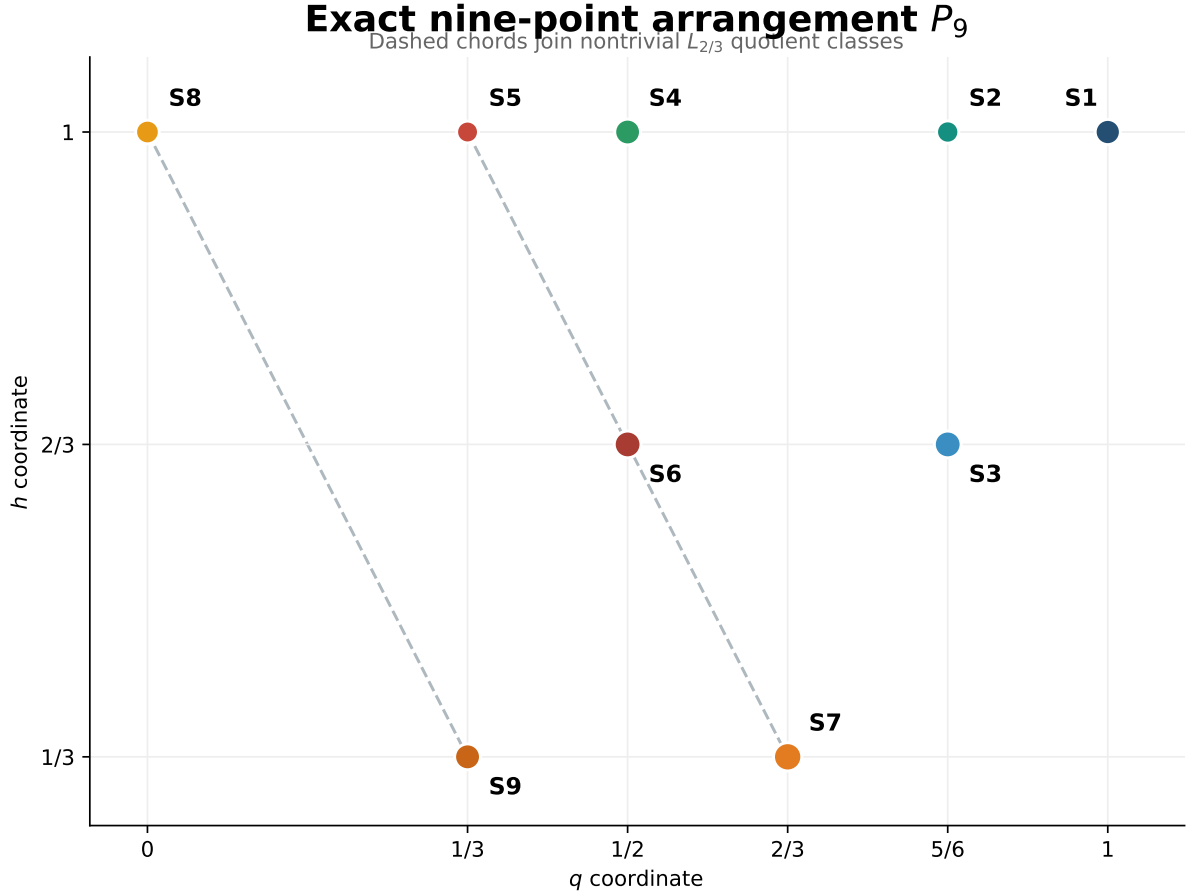


Figure 1. The independent rational arrangement P_9 in the (q, h) plane. Sector labels identify the nine declared points. Dashed chords mark the nontrivial $L_{2/3}$ quotient classes of this exact finite arrangement; the figure does not certify an exact Rubik joint spectrum.

Theorem 5.1 (Exact nine-point census).

For the displayed rational arrangement:

1. the 36 unordered pairs split into 2 parallel, 10 interior, 15 endpoint, and 9 exterior pairs;
2. the interior collision set on $[0, 1]$ is

$$\{2/7, 2/5, 1/2, 2/3, 4/5\};$$

3. the critical quotient data are:

α	Classes	$N_P(\alpha)$	$\Delta_P(\alpha)$	$m_{\text{pair}}(\alpha)$
0	S1=S2=S4=S5=S8; S3=S6; S7=S9	3	6	12
2/7	S3=S8	8	1	1
2/5	S3=S5; S6=S8	7	2	2
1/2	S3=S4; S7=S8	7	2	2
2/3	S5=S6=S7; S8=S9	6	3	4
4/5	S4=S7	8	1	1
1	S2=S3; S4=S6; S5=S9	6	3	3

4. $\alpha = 2/3$ is the unique **interior** parameter maximizing the layer-count drop Δ_P .

Proof. Appendix [Appendix A](#) lists all $\binom{9}{2} = 36$ pairs, their exact α_{ij} values, and their classifications. The table contains 2 parallel, 10 interior, 15 endpoint, and 9 exterior pairs. Grouping equal interior values yields $\{2/7, 2/5, 1/2, 2/3, 4/5\}$. Direct substitution of the nine displayed points at the seven parameters in the critical table produces the quotient classes, sizes, drops, and pair multiplicities. The interior drops are 1, 2, 2, 3, 1, so the interior maximum is uniquely attained at $2/3$. Every step uses only the displayed rational coordinates and the collision formula.

Corollary 5.2 (Exact weighted quotient at alpha two-thirds).

At $\alpha = 2/3$, the weighted quotient of P_9 is

Value	Members	Total weight
1	S1	20
8/9	S2	2
7/9	S3	39
2/3	S4	26
5/9	S5, S6, S7	106
1/3	S8, S9	35

This is an exact statement about the displayed weighted point set.

5.1 Chamber order on the unit interval

Every open chamber has nine distinct values. The descending branch orders are:

Chamber	Descending order
$(0, 2/7)$	S1, S2, S4, S5, S8, S3, S6, S7, S9
$(2/7, 2/5)$	S1, S2, S4, S5, S3, S8, S6, S7, S9
$(2/5, 1/2)$	S1, S2, S4, S3, S5, S6, S8, S7, S9
$(1/2, 2/3)$	S1, S2, S3, S4, S5, S6, S7, S8, S9
$(2/3, 4/5)$	S1, S2, S3, S4, S7, S6, S5, S9, S8
$(4/5, 1)$	S1, S2, S3, S7, S4, S6, S5, S9, S8

Together with the critical table, this is the full interpolation phase diagram of P_9 on $[0, 1]$.

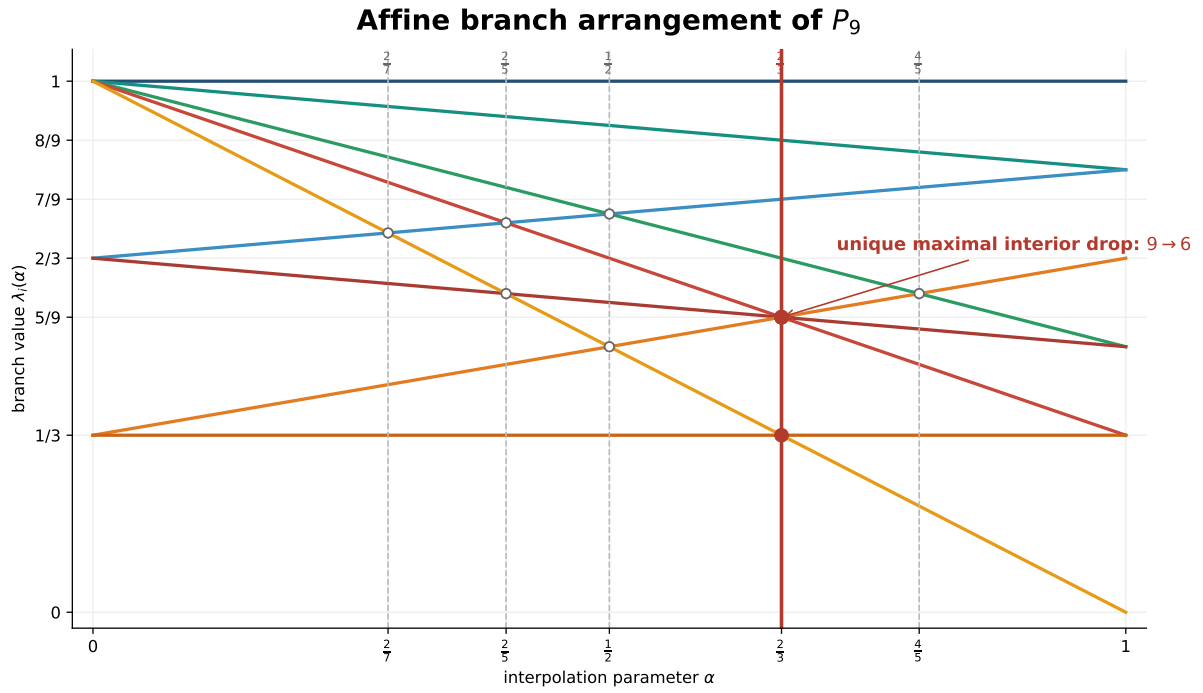


Figure 2. Affine branches of the independent arrangement P_9 . The five interior critical parameters and the unique interior maximum of the layer-count drop are exact finite-point statements. Their identification with an actual Rubik joint spectrum is conditional on the registration hypotheses below.

Interior collision graph of P_9

Red edges occur at $\alpha = 2/3$; gray edges mark the other interior collisions

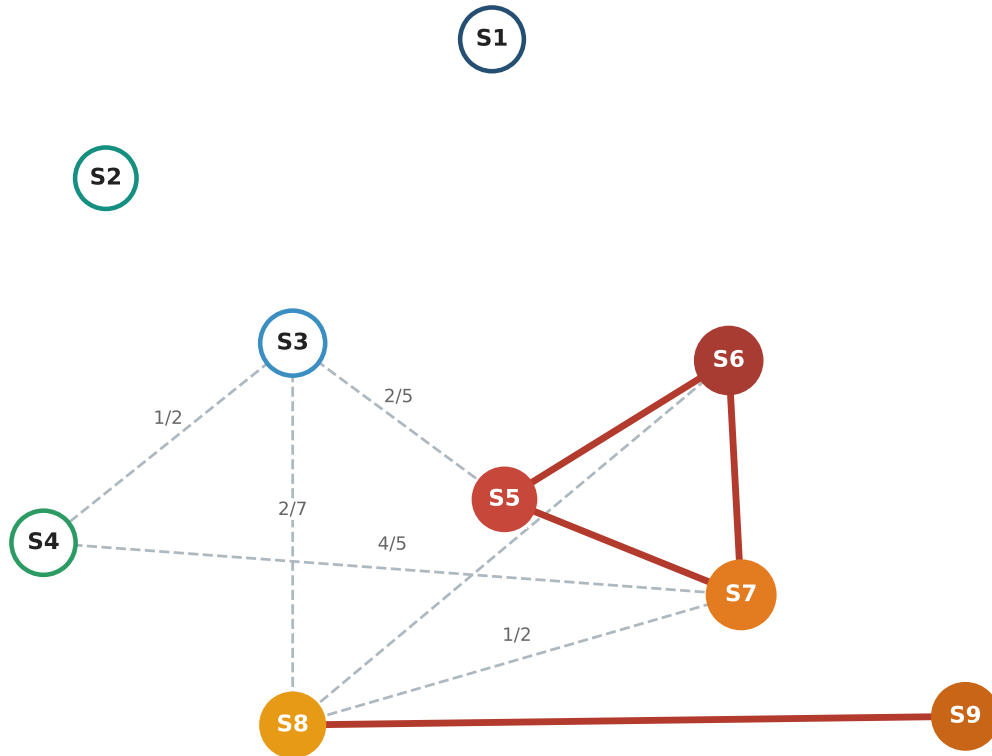


Figure 3. Interior collision graph of P_9 . Red edges have label $2/3$ and form the exact triple class $S5=S6=S7$ together with $S8=S9$. Gray dashed edges mark the other exact interior collisions of the displayed arrangement.

6 Computational Rubik Registration

In the declared 228-dimensional complex128 cubie realization, define

$$Q_{\text{num}} = QT_{\text{all}}, \quad H_{\text{num}} = HT_{\text{all}}.$$

The generator counts give the definitional identity

$$A_{18} = \frac{12Q_{\text{num}} + 6H_{\text{num}}}{18} = \frac{2}{3}Q_{\text{num}} + \frac{1}{3}H_{\text{num}}.$$

This identity does not imply that Q_{num} and H_{num} commute exactly or that their numerical clusters have exact rational eigenvalues.

6.1 Registration algorithm

For each declared clustering tolerance τ , the registration is the following deterministic numerical procedure. First diagonalize Q_{num} and partition its ordered numerical eigenvalues into clusters by the absolute rule

$$|\hat{q}_j - \hat{q}_i| < \tau,$$

where $\hat{q}_i$ is the first eigenvalue in the cluster. Within each resulting Q_{num} eigenspace, diagonalize the restricted H_{num} and apply the same absolute rule. The resulting orthonormal basis B_i defines

$$P_i = B_i B_i^*, \quad \hat{q}_i = \frac{\text{tr}(B_i^* Q_{\text{num}} B_i)}{\dim B_i}, \quad \hat{h}_i = \frac{\text{tr}(B_i^* H_{\text{num}} B_i)}{\dim B_i}.$$

Thus the representative is a mean Rayleigh scalar and the rank is the number of orthonormal basis columns. The nine raw coordinate pairs are then matched one-to-one to the predeclared set P_9 by a Hungarian assignment with cost

$$d_{\infty}((q, h), (q', h')) = \max\{|q - q'|, |h - h'|\}.$$

There is no random step, denominator search, or rational reconstruction: the procedure measures only agreement with the independently declared table.

Proposition 6.1 (Registered QT/HT clusters; computational).

In the declared complex128 construction, the registration algorithm above produces nine clusters with coordinate-matched ranks

$$(20, 2, 39, 26, 1, 39, 66, 8, 27).$$

Their raw scalar coordinates match the rational coordinates of P_9 . The maximum raw-to-table discrepancy across the QT, HT, and A_{18} coordinates is

$$9.992 \times 10^{-16}.$$

The principal numerical diagnostics are:

Diagnostic	Frobenius residual
$Q_{\text{num}} - Q_{\text{num}}^*$	2.73×10^{-16}
$H_{\text{num}} - H_{\text{num}}^*$	0
$[Q_{\text{num}}, Q_{\text{num}}^*]$	3.17×10^{-18}
$[H_{\text{num}}, H_{\text{num}}^*]$	0
$[Q_{\text{num}}, H_{\text{num}}]$	3.77×10^{-16}
$A_{18} - (2Q_{\text{num}} + H_{\text{num}})/3$	2.94×10^{-17}
$\max_i \ P_i^2 - P_i\ _F$	1.087×10^{-14}
$\max_i \ P_i - P_i^*\ _F$	6.314×10^{-16}
$\max_{i < j} \ P_i P_j\ _F$	4.782×10^{-15}
$\ \sum_i P_i - I\ _F$	2.229×10^{-14}
maximum rank–trace discrepancy	1.421×10^{-14}
maximum joint-eigen residual	8.661×10^{-15}

For $\tau \in \{10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}\}$, the audit returns nine clusters with the same coordinate-matched label–rank pairs, zero coordinate and projector drift relative to $\tau = 10^{-10}$ at printed precision, and minimum normalized trace overlap

$$\min_i \frac{\text{tr}(P_i(\tau)P_i(10^{-10}))}{\text{rank } P_i} = 0.9999999999999983.$$

The predeclared arrangement has the exact separation

$$\min_{i \neq j} \|p_i - p_j\|_\infty = \frac{1}{6}.$$

Appendix [Appendix B](#) records the complete tolerance certificate.

Status. This is computational evidence. It certifies numerical stability and proximity to the declared rational table, not exact commutation, exact rationality, or exact primitive idempotents.

Within this numerical layer, S1–S9 are called the **registered joint clusters**. The phrase “primitive idempotent of $\mathbb{C}[Q, H]$ ” is reserved for the exact conditional setting.

7 Conditional Rubik Interpretation

Corollary 7.1 (Conditional Rubik collision quotient).

Assume that:

1. the exact Rubik operators $Q = QT_{\text{all}}$ and $H = HT_{\text{all}}$ are commuting Hermitian operators;
2. their exact joint eigenspaces have ranks and joint eigenvalue coordinates given by P_9 ;
3. $A_{18} = (2Q + H)/3$.

Then the exact A_{18} eigenspaces are the $L_{2/3}$ collision classes of P_9 . In particular, their eigenvalues and dimensions are

$$(1, 20), (8/9, 2), (7/9, 39), (2/3, 26), (5/9, 106), (1/3, 35).$$

Proof. Apply Theorem [3.2](#) and Corollary [5.2](#).

Under these hypotheses, the nine exact joint projectors are the primitive idempotents of the represented algebra $\mathbb{C}[Q, H]$. This primitivity is relative to the declared algebra only. It does not assert maximality among larger commuting extensions in $\text{End}(V)$.

The corollary explains the relationship between the registered six-layer census and the finer nine-cluster table. However, it does not prove the exact registration hypotheses, nor does it provide an independent arithmetic proof of canonical Rubik rationality. The corresponding six-layer census has been recorded elsewhere as a finite computation with mixed analytic and numerical input [3].

8 Comparison with Transport

Collision adjacency and transport adjacency are different objects. For the explicit arrangement, the 2/3 collision graph contains the complete triangle

$$S5-S6, \quad S5-S7, \quad S6-S7.$$

8.1 Triangle versus chain

Computational observation.

In the declared Rubik numerical registration, direct generator-support blocks are nonzero for $S5-S6$ and $S6-S7$ and below the declared threshold for $S5-S7$. Thus the registered direct-support graph on these three clusters is the chain

$$S5-S6-S7,$$

not the collision triangle.

This observation does not imply that collision geometry determines transport. Collision geometry records equality of affine scalar values, whereas direct support records nonzero projected generator blocks. Projected composition requires a further image-kernel audit [4]. If sectors are merged, graph incidence and operator products must be explicitly recomputed at the new resolution rather than automatically pushed forward through the collision quotient.

9 Claim Status and Boundary

The table uses the four claim levels. Exact finite theorems and conditional corollaries remain in the Theorem level under their displayed hypotheses.

Claim	Status
Collision formula and finite critical set	Theorem
Chamberwise constancy of branch order	Theorem
Spectral layers equal collision classes for commuting Hermitian Q, H	Theorem
Rational coordinates imply rational finite collision parameters	Theorem
Complete census of the displayed P_9	Theorem
2/3 is the unique interior maximal-drop point of P_9	Theorem
Numerical QT/HT clusters match P_9 to declared residuals	Computational Certificate
Actual Rubik layers are the exact 2/3 quotient, conditional on exact registration	Theorem
Collision triangle differs from direct-support chain	Computational Observation
QH algebra is canonical or maximal among commuting refinements	Research Program
Moving arrangements and spectral-wall inclusions	Research Program
Spectral coarse-graining preserves information flow	Research Program

The principal failure boundary is the promotion step:

$$\text{numerical cluster registration} \not\Rightarrow \text{exact joint spectral theorem.}$$

Exact arithmetic on the predeclared table validates that table. It does not retroactively prove that the numerical operator data are exact rational spectral data.

10 Related Work and Novelty Boundary

Simultaneous diagonalization and joint eigenspaces are standard matrix analysis [8]. This fixed commuting setting differs from general Hermitian pencils and analytic perturbation theory: eigenvalue branches need not be affine, and noncommuting perturbations can produce avoided crossings. Classical perturbation theory supplies that broader boundary; see [10] and [13].

Pair equalities

$$L_\alpha(p_i - p_j) = 0$$

are one-dimensional sections of a hyperplane discriminant arrangement. The language of hyperplane arrangements and discriminants provides the natural geometric comparison class [11]. This paper does not claim a new general eigenvalue-perturbation theorem. Its contribution is the explicit organization of a fixed joint-spectrum projection as a collision quotient, the exact census of P_9 , and the claim-status-separated Rubik registration.

Association schemes provide a second comparison class: their Bose–Mesner algebras are finite-dimensional commutative semisimple algebras organized by primitive idempotents, and quotient schemes arise from additional equitable partition structure [1, 5, 6]. No association-scheme or coherent-configuration structure is claimed for the registered QT/HT pair. Godsil–Martin quotient theory provides a comparison analogy, not a proof of Theorem 3.2.

The Rubik operators are represented group-algebra averages in the standard finite-group and cubie setting [12, 9]. Related studies record the block-spectral census, direct transport on the registered QT/HT clusters, and the separation between support-graph paths and projected matrix composition [3, 2, 4]. These results are not used in the general finite-point proofs.

11 Research Program

The following directions are not established in this article.

11.1 Moving arrangements

Generator deletion or reweighting may replace the fixed arrangement P by a moving family $P(w)$. Before collision quotients can be assigned, a deformation theory must pass the gates

$$\begin{array}{c} \text{commutativity} \longrightarrow \text{normality} \longrightarrow \text{spectral chart} \\ \downarrow \\ \text{orthogonal joint projectors} \longrightarrow \text{candidate layer and field walls} \end{array}$$

This sequence remains a proposed programmatic outline. The existence of such charts, constancy of rank, projector continuation, and wall inclusions are not established here. Commutativity alone does not supply normality or orthogonal joint projectors.

11.2 Candidate wall hierarchy

Symbols such as Σ_{comm} , Σ_{spec} , Σ_L , and Σ_{field} belong to a deformation program. No subset chain among them is asserted here. Establishing a valid hierarchy would require definitions on a common admissible weight space and proofs of each domain inclusion.

11.3 Maximal commuting refinements

The numerical QH clusters are minimal at the declared clustering tolerance. Under the exact hypotheses of Corollary 7.1, the exact joint projectors would be primitive relative to $\mathbb{C}[Q, H]$. Whether this algebra is canonical or maximal in a symmetry-constrained class is open.

11.4 Higher-dimensional arrangements

For $P \subset \mathbb{R}^d$ and a weight vector w , pair collisions lie on the hyperplanes $w \cdot (p_i - p_j) = 0$. Developing the corresponding chamber and quotient theory beyond the one-dimensional interpolation studied here is a natural extension.

11.5 Spectral coarse-graining and information flow

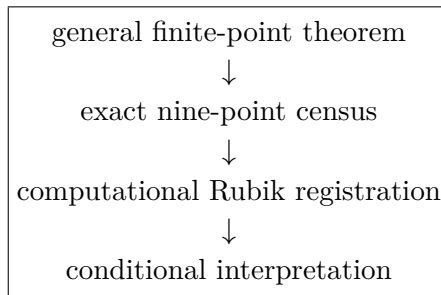
The triangle-versus-chain observation shows that a spectral quotient can hide resolved direct-support data. A future spectral coarse-graining or “spectral RG” theory would need explicit criteria for which support, word-composition, or Lie-accessibility data survive quotienting. While the Rubik triangle-versus-chain comparison motivates this question, it does not establish a general information-flow theorem.

12 Conclusion

For a fixed finite arrangement, collision quotients are exact finite objects. The critical set is finite, the branch order is chamberwise constant, and a commuting-Hermitian realization identifies quotient classes with spectral layers. The explicit rational set P_9 admits a complete exact census on $[0, 1]$, with $2/3$ the unique interior parameter of maximal layer-count drop.

The Rubik interpretation has a different status. The declared numerical operators produce nine stable clusters extremely close to P_9 . However, numerical residuals and table agreement do not prove exact joint spectral data. Conditional on that exact promotion, the six A_{18} layers are the $L_{2/3}$ collision quotient of the nine-point arrangement.

The resulting architecture is therefore:



Moving arrangements, normal spectral charts, maximal refinements, and information-flow preservation begin only after this fixed-arrangement theory.

Appendix A Complete Pair Certificate

This appendix gives a human-checkable certificate for Theorem 5.1. For each unordered pair, the entry is obtained by substituting the displayed rational coordinates into Theorem 3.1. A dash means equal slopes and hence no finite collision. The four tables contain all 36 pairs exactly once.

A.1 Parallel pairs

Pair	α_{ij}	Classification
S1–S9	–	parallel
S2–S6	–	parallel

A.2 Interior pairs

Pair	α_{ij}	Classification
S3–S4	1/2	interior
S3–S5	2/5	interior
S3–S8	2/7	interior
S4–S7	4/5	interior
S5–S6	2/3	interior
S5–S7	2/3	interior
S6–S7	2/3	interior
S6–S8	2/5	interior
S7–S8	1/2	interior
S8–S9	2/3	interior

A.3 Endpoint pairs

Pair	α_{ij}	Classification
S1–S2	0	endpoint
S1–S4	0	endpoint
S1–S5	0	endpoint
S1–S8	0	endpoint
S2–S3	1	endpoint
S2–S4	0	endpoint
S2–S5	0	endpoint
S2–S8	0	endpoint
S3–S6	0	endpoint
S4–S5	0	endpoint
S4–S6	1	endpoint
S4–S8	0	endpoint
S5–S8	0	endpoint
S5–S9	1	endpoint
S7–S9	0	endpoint

A.4 Exterior pairs

Pair	α_{ij}	Classification
S1–S3	2	exterior
S1–S6	−2	exterior
S1–S7	2	exterior
S2–S7	4/3	exterior
S2–S9	4	exterior
S3–S7	2	exterior
S3–S9	−2	exterior
S4–S9	4/3	exterior
S6–S9	2	exterior

The counts are therefore $2 + 10 + 15 + 9 = 36$. Grouping the ten interior rows by their exact parameter gives multiplicities 1, 2, 2, 4, 1 at $2/7, 2/5, 1/2, 2/3, 4/5$, respectively. The quotient-class table in Theorem 5.1 then follows by transitive closure at each common parameter.

Appendix B Registration Stability Certificate

The baseline registration uses $\tau = 10^{-10}$. Across all four tested tolerances, coordinate matching to P_9 returns the same label–rank list

$$(S1, 20), (S2, 2), (S3, 39), (S4, 26), (S5, 1), (S6, 39), (S7, 66), (S8, 8), (S9, 27).$$

The projector comparison uses the labels assigned by the global d_∞ matching, not a sorted rank census.

τ	Clusters	Label–rank stable	Max d_∞ match to P_9	Max coordinate drift
10^{-6}	9	yes	9.992×10^{-16}	0
10^{-8}	9	yes	9.992×10^{-16}	0
10^{-10}	9	yes	9.992×10^{-16}	0
10^{-12}	9	yes	9.992×10^{-16}	0

τ	Max projector drift in Frobenius norm	Min normalized overlap
10^{-6}	0	0.9999999999999983
10^{-8}	0	0.9999999999999983
10^{-10}	0	0.9999999999999983
10^{-12}	0	0.9999999999999983

Zeros denote values returned in floating-point complex128 arithmetic, not exact symbolic identities. The overlap is

$$\frac{\text{tr}(P_i(\tau)P_i(10^{-10}))}{\text{rank } P_i}.$$

The implementation records the clustering rule, matching metric, dtype, package versions, source hashes, Git state, and runtime in the observation artifact. These data certify reproducibility of the stated numerical audit; they do not promote numerical projectors or coordinates to exact objects.

Appendix C Computational Artifacts

The following repository artifacts support the exact finite census and the computational Rubik case study. The default directory is `experiments/paper4/`; paths are relative to that directory.

Artifact	Role	Short path
C1	exact Fraction census of the declared P_9 arrangement	<code>validation/rubik_collision_quotient.py</code>
C2	numerical Rubik registration and residual audit	<code>validation/rubik_joint_spectrum_registration.py</code>
C3	collision-triangle/direct-support-chain comparison	<code>validation/v59_collision_vs_transport.py</code>
C4	source-addressed registration observation	<code>results/rubik_joint_spectrum_registration_observation.json</code>

C2 reports operator Hermiticity and normality, QT/HT commutation, A_{18} reconstruction, projector identities, ranks, joint-eigen residuals, raw-to- P_9 discrepancies, ℓ_∞ separation, and matched-projector stability. From the repository root, run an executable artifact as `python experiments/paper4/<short path>`; append `--check-result` to C2 to check C4 against its declared source hashes.

C4 is a source-addressed run record, not an independent proof. Reproducing the certificate requires a full run of C2.

All listed artifacts are available in the [RIME repository](#).

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